

# **FUNDAMENTALS OF MAINTENANCE ENGINEERING**

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## 1.0 Introduction :

Efficient industry is coming up as a result of frequent new inventions, use of modern machineries, proper planning and producing quality products. Industrial development is the measure for the strength of the nation and quality products is the base of industrial development. To fulfil this basic need, the use of efficient machineries is unavoidable. The branch of engineering which keeps machines always in efficient condition. The engineering branch which gives the knowledge of keeping the machineries and utility always in good & efficient condition is known as maintenance engineering.

The machines are considered of prime importance due to fast industrial development and advancement in technology. In past when machine met with breakdown due to defective running & stops functioning than & than the maintenance effort was carried out. But now-a-days breakdowns are considered as unwanted in modern industries. As the breakdown of machines leads to stoppage of production, needs time consuming and costly repairing & replacement of parts. In the events of stoppage of production the income stops and cost of repairing leads to increase in expenses, which finally gives higher cost of production. In the industrial field there prevails tough competition amongst the industries, therefore to service in this field it is unavoidable to adopt the maintenance of a kind, which may replace the breakdown maintenance. Therefore as per the demand of time to produce desired quantity of products keeping its quality intact preventive maintenance is now-a-days carried out to keep machines & utilities in accurate condition, which leads to their increased productivity.

Due to friction, the parts of machines suffers wear and tear, when parts gets wornout their dimensions changes, which alters the clearance between the pair of machines parts. The changes of clearance results into the alteration of adjustments made and hence machines loses their capacity to give accuracy as per the demands of the customers. To reduce the wear & tear the main aid is the lubrication. Timely lubrication, regular cleaning and adjusting the changed clearances can keep the machines in efficient condition for longer period of time maintaining the condition of machines and using them in efficient condition, the quality productions can be obtained. Therefore as per the scientific ways are adopted in newly designed preventive maintenance to maintain & repair the machines to produce the quality products. Such new method of maintenance have successfully replaced the old method of breakdown maintenance. The periodical and planned efforts of maintenance & repairing are implemented in preventive maintenance programme for the machineries. In this type of maintenance methodical care is taken for each and every machines & utilities. Maintenance and repairing schedules are prepared keeping in mind the importance of work and each machine. For that in depth planning is carried out maintenance and repairing work and put up into actual practice. Therefore the field developed based on new & scientific ideas is known as "maintenance engineering."

In maintenance of equipments & machines, over and above removal of the fault, regular oiling, greasing, adjustments & repairing is also carried out to prevent the fault developing.



Therefore the field of maintenance & prevention of faults in the machineries by keeping them in good condition has been given a name of "maintenance engineering."

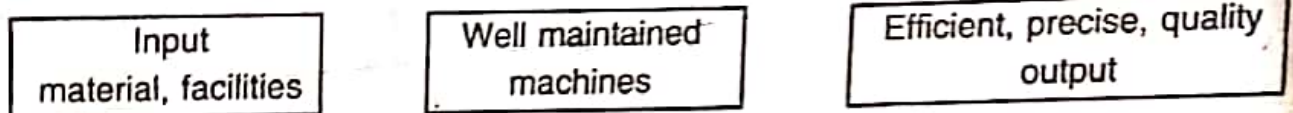
Maintenance engineering gives the guidelines to keep the machineries & equipments always in good & efficient condition. It also suggests the ways & means for the implementation of guidelines. Hence preventive maintenance is of utmost importance in modern industries to achieve the goal of producing products in right quantity of desired quality at right time with the reduced cost of production to enter into the healthy competition in modern markets to sell the products.

## 1.1 Definition and Aims of Maintenance Engineering :

### 1.1.1 Definition of Maintenance :

(4)

To keep the machines, plants & process equipments in their efficient conditions, the maintenance activities are planned and carried out is known as "maintenance". This results into the reduced downtime of the machines and gives accurate and best quality of products produced by using the various facilities in an industry.



### 1.1.2 Maintenance Engineering :

Maintenance Engineering is a technical field which gives us the knowledge of maintenance and ideas for their implementation to keep the industrial machines, plants & processes in efficient condition in which latest methods of maintenance are developed by timely bringing the necessary changes based on scientific development. This is the field which maintains the accuracy & quality of the products by maintenance of industrial machines and other facilities.

### 1.2.3 Aims of Maintenance Engineering :

1. To maintain the machines & equipments to obtain quality products as well as to maintain the productivity.
2. To prevent sudden and incidental breakdown of machines.
3. To achieve safe utilization of machines.
4. To overhaul the machines as per schedule to make them efficient.
5. To obtain maximum efficiency to machines by maintaining them.
6. To reduce the downtime of machines which became out of production.
7. To reduce the cost of maintenance.
8. To exercise the maintenance activities as per higher engineering standards.
9. To prevent or reduce the production wastage.
10. To reduce cost of production by bringing economy in maintenance.
11. To maintain the position of industry in these days of competitive age.



12. To carry out every maintenance activity with the aid of both technical knowledge and experiences.
13. To increase the service life of plants & machineries.
14. To maintain the record of maintenance work for future guidelines.
15. To provide in time infrastructural facilities to the industrial unit.
16. To prevent the occurrence of accidents and to bring safety in the industry.
17. To reduce the loss to the industry owing to the accidental fire.
18. To erect the new plants, machines & equipments systematically.
19. To maintain and manage the store for maintenance requirements.
20. To help in maintaining or to increase the sale of product by bringing productivity & quality in production.
21. To achieve economical stability of the industry.
22. To prepare and implement the preventive maintenance schedules.
23. To carry out maintenance, repair & replacement periodically.

## **1.2 Functions and Responsibilities of Maintenance Department :**

The functions of maintenance departments are divided in two groups. These groups are named as primary functions and secondary functions. The detailed explanation of the functions included in both these groups are as under :

### **1.2.1 Primary Functions : (12)**

The primary functions of maintenance department are :

1. Maintenance of existing plant and machineries.
2. Inspection and lubrication of equipments.
3. Proper distribution of utility services.
4. Modification or alteration of existing equipments.
5. Installation of new equipments & construction of building.
6. Maintenance of existing building & ground.

#### **1.2.1.1 Maintenance of Existing Plant and Equipments :**

This is the main function of the maintenance department as this department is came into existence by the virtue of this function only. Therefore to satisfy the needs of this function following activities are carried out by the maintenance department in an industry.

1. To carry out speedy & economical repairing of production machineries.
2. To prevent repairing work as far as possible using proper preventive maintenance but at the same time to keep ready the informations for completing such repairing work in shortest possible time.
3. To develop a team of trained and skilled workers for the maintenance activities.
4. To reduce both the down time and maintenance time.
5. To make it possible to distribute maintenance expenses by keeping suitable records.



6. To prepare & implement maintenance planning & schedules.
7. To plan and carry out maintenance work with a view to reduce the labour-cost.
8. The replacement parts and other required materials are to be stored as per economic quantity.
9. To fix strong and sufficient guards to the machines for preventing accidents.
10. To draw preventive maintenance programme and to implement the same in the industry with utmost care.
11. To remove the reasons and defects which are responsible for deterioration of machines.

#### **1.2.1.2 Inspection and lubrication of equipments :**

This function is carried out systematically by the maintenance department in medium and heavy industries. Maintenance work is carried out after inspections in this function attempt is made to keep machinery in good condition by regular oiling, greasing and changing the oil in reservoir of the machines at proper time. The purpose of the lubrication is satisfied by preparing & implementing the oil changing programmes. In maintenance the meaning of the inspection is not limited only for checking or visualize. Here the meaning of inspection is to carry out the important work for maintenance, which needs very little time. In this inspection activity following works are carried out :

1. Cleaning of the machines.
2. To adjust the clearance and required setting of wornout parts.
3. To tighten the loose nut-bolts.
4. To bring the safety by proper fixing of machine-guards.
5. To reduce the friction between wearing parts by proper lubrication.
6. To reduce wear & tear of the parts and to increase the service-life of the machines by regular attention & lubrication.

#### **1.2.1.3 Proper distribution of utility services :**

The heavy industrial units have their own powerplant, where electricity is produced. Moreover industry needs air, water, steam etc. in processes. All these needs are named utility services. To produce and distribute such utilities is the function of maintenance department. To discharge this function the department is producing electricity, steam, compressed air and supply required quantity of water. For to serve this to the needy sections of the industry, the maintenance department manages the power-house, boiler house, compressor and pump house. These utilities services are the basic needs of the production processes, therefore, proper quantity of the same at required time are supplied by the maintenance department considering this work as an important work of the department itself.

#### **1.2.1.4 Modification or Alteration of Existing Equipments :**

This is the work of maintenance department for up-dating the existing machineries. As per the developed technology changes are brought about in the machineries with a view to



improve their condition. It increases the efficiency and productivity of the machines as well as good quality products can be produced by using these improved machineries. The modernization carried out to bring the ease in production. To grasp the work from old machineries same as the newly developed machines, the improvement carried out base on developed technology is known as modification or alteration of machines, which serves as a tool to make older. Machines improvement to make them equally capable in comparison to newer machines.

#### 1.2.1.5 Installation of new equipments and construction of building :

Erection of various equipments is the work assigned to the maintenance department in the industry owing the size of the plant, more than one plant in an industry and as per the policy laid down by the management. Maintenance department is suppose to carry out the new constructional work for the industry as well as to carry out the constructional work to modify the existing building over and above the work of erection of the maintenance and equipments.

#### 1.2.1.6 Maintenance of existing building and ground :

Maintenance department is looking after work of maintaining the roads, Railway tracks, sewers and water supply and drainage lines in an industry. The building & grounds needs to be cleaned regularly. In some of the industrial units these types of work are carried out by the material handling department. Even separate department is created in industry when the size of plant and building is huge to look after such work. Eventhough repairing, whitewash & oil paint work, electrical & plumbing work, maintenance of doors & windows etc. are carried out by the maintenance department. On the same way maintenance of roads, rails, sewer, water pipelines, drainage lines etc. are also carried out by the maintenance department.

Maintenance department has the specific – work carrying responsibility as per the size, type and policy of the management of the plant. Fast industrial development has resulted into variety of plant, each having different requirements of work. Therefore due to diversification of industrial plants, the responsibility of the maintenance departments & job assigned to the department are different in the different plants. The responsibility given to the maintenance department is according to the suitability of the individual plant in each case. The modern needs of productivity, production and its quality requirement has compelled the different plants to use different types of automatic & semi automatic machines leading to the varied job requirement of the maintenance department.

#### 1.2.2 Secondary Functions :

The secondary functions of the maintenance department are :—

1. Protection of the plant.
2. Waste disposal of the plant
3. Salvage
4. Maintenance of store



5. Insurance – administration
6. Property accounting
7. Pollution and noise – control.
8. Sanitation & service facility administration
9. Other services.

#### **1.2.2.1 Protection of the plant :**

Security management & fire protection management staff of the maintenance department are provided for protecting the plant. Protection of plant is an activity does not carry out by the maintenance department in each & every plant. As per the local condition of the plant this responsibility is given to the maintenance department. Certain plants have their separate security department. Safeguard against theft and damage prevention due to fire etc. are protected by the security department. By checking the workers, visitors and goods or materials coming in and going out, security department prevents the occurrence of loss to the plant. To protect the plant against fire, fire protection and fire fighting equipments are kept always in ready condition in sufficient numbers. Over and above the workers are trained by providing them through informations as regards the use of each equipment meant for fire extinguishing work.

#### **1.2.2.2 Waste disposal of the plant :**

Unserviceable parts, broken furnitures, electrical unserviceable materials and scrap of production processes etc. are grouped in different groups. Each group has been assigned a separate lot numbers and arranged & kept separately in the scrap-yard of the plant. Record related to scrap is prepared. In this record lot number, description of scrap, approximate quantity etc. are recorded. Each lots are auctioned and handed over to the purchasers after recovering the auction price from them.

#### **1.2.2.3 Salvage :**

Further deterioration of the machine parts, machines, containers etc. is prevented by arranging them in lots and timely disposal of the same. The upset price of each item is calculated and fixed by deducting the depreciation from total price at the end of service life. But such items are sold at a higher price in comparison to their scrap-value. The selling is decided at a price based on its condition which is known as salvage price. Periodical sale of such items keeps the scrap yard ready to accommodate another scrap as well as the money realized as per the salvage price can be used in other useful purpose. This secondary responsibility of maintenance department needs prompt and timely attention.

#### **1.2.2.4 Maintenance of store :**

Store has two sections namely mechanical store & general store. Maintenance department needs different replacement items of both types of stores at the time of repairing. The number of each store items used in a machine, number of machines, approximate service life of the item etc. are the base for deciding the stocks to be kept in store for each item. This stock is



maintained in such a way that the items for replacement is available as & when needed as well as the capital requirement for maintaining the stock should also be minimum. To maintain the stock, regular purchasing is done. The account of incoming store and consumed store is also kept. Store receipt, issue and storage procedures are laid down and followed for proper accounting. Each items are entered in store ledgers and annual inventory of the stock is carried out in this secondary function by the maintenance department.

#### **1.2.2.5 Insurance – administration :**

To prevent the economical hardship arises due to accidental damage to the building, machineries and platnt, they are covered by the proper insurance policies. Insurance premiums are paid in time & regularly. When such type of damage occurs than to claim the losses, preparation of claim documents and after submitting the claims to the Insurance company, to get the compensation of loss from them is also a secondary function of the maintenance department.

#### **1.2.2.6 Property accounting :**

This is the work of keeping records in stock-ledger for the machineries, equipments, building and vehicles giving their detailed description. This record is maintained to get the minute informations of cost, place and condition of the different properties of the plant. This property accounting is a secondary function of the maintenance department.

#### **1.2.2.7 Pollution and noise control :**

The efficiency of the workers in the plant reduces to the pollution and noise. This pollution and noise are also responsible for the deteriorating health of the workers on the long run. Smoke and poisonous gas leakage can create hazardous atmosphere in the plant and cause injury to the workers. Therefore the work of controlling pollution and noise level is also assigned to the maintenance department as its secondary function. The smoke, poisonous gas leakage, smell of burning oil, combustion of rubber and plastic, waste water are the certain examples of pollution, which are controlled by the maintenance department.

#### **1.2.2.8 Senitorial & Service facility administration :**

Floor, window glass cleaning and washing, repairing of doors & windows etc. are the work assigned to the maintenance department, looking to the volume of work and management policy. The utility services as related to the steam, electricity, water supply etc. are run by the maintenance department is also a secondary function.

#### **1.2.2.9 Other services :**

The work which are not assigned to another department or the other department is not ready to carry out a particular work than that type of work is assigned as an additional work to the maintenance department. Construction supervision & production consultation are some of the examples of additional work. To carry out such secondary functions satisfactorily and to maintain good relations with other departments, certain powers and responsibilities are fixed by the management of the plant for the maintenance department.

**1.2.3 Responsibilities of maintenance department :** (6)

The responsibilities of the maintenance department are as follows :

1. Planning & implementation of preventive maintenance, periodic maintenance, repairing and replacement work.
2. Protection against fire hazards & fire extinguishing
3. Breakdown maintenance
4. To provide guidelines & suggest alteration, modification & replacement etc..
5. Maintenance of store.
6. To maintain the standard of maintenance work.
7. To supervise the activities of maintenance undertaken in the plant.
8. To suggest incentive for the staff of maintenance department.
9. To train the maintenance department staff.
10. To keep the record of maintenance work already carried out.
11. To keep the machineries & equipments in good & efficient condition.
12. To provide utility services like steam, electricity, water etc.
13. To keep the property accounts.
14. To control the pollution and noise.
15. To carry out work related to other services and maintain good relations with the other departments of the plant.
16. To look after the insurance administration.
17. To undertake the work of waste disposal of the plant.
18. To fix the upset price of scrap machines, parts & other items so that their selling price can be realized according to their conditions.
19. To prevent breakdowns and bring the safety amongst the plant.
20. To reduce the cost of maintenance.
21. To increase the service-life of the plant, machinery & equipments.
22. To prevent or reduce the production wastage.
23. To bring economical stability of the plant.

**1.3 Types of Maintenance :**

The different types of maintenance are as under :

1. Capital replacement
2. Breakdown maintenance
3. Corrective maintenance
4. Scheduled maintenance
5. Planned maintenance
6. Productive maintenance



7. Preventive maintenance
8. Predictive maintenance
9. Modular maintenance

### 1.3.1 Capital replacement :

In capital replacement type of maintenance, the replacement of the faulty units, assemblies and electric-motors are replaced instead of repairing them. To crate the facility for replacement spare units assemblies and electric motors are kept ready in the store. This type of maintenance needs lesser time, hence down time of machine reduces as a result of which the machine can be put in use in less time. This type of maintenance does not require any engineering analysis, maintenance planning etc. therefore it is not necessary to keep any record for the future but it is costly is the disadvantage of it, hence it is adopted when plant has sound economical background. In this system the replacement is done without recovering the cost behind the replacement items.

To understand the capital replacement consider an example of a automobile truck. Its gearbox is found faulty. The repairing of the truck is carried out by replacing the faulty gearbox by another gear-box readily available in the store. Due to lesser time utilized or taken in such work the reduction of down time due to this type of practice results into no effect on transportation income. In case if gear box was dismantled, fault finding was carried out & gear box was repaired after replacing the faulty part than to fit the gearbox in the truck may have consumed more time, resulting into loss of transportation income.

### 1.3.2 Breakdown maintenance : (16)

In this type of maintenance, the attention is given to the machine when it stops functioning. To repair the machine fault finding is done which consumes more time. After locating the faulty part, the repairing or replacement of the part is done to repair the machine. Therefore down time in this type of maintenance is too high, which hampers the production due to which target of production cannot be achieved. Therefore breakdown maintenance is undesirable practice in the plants.

#### 1.3.2.1 Advantages of breakdown maintenance : (16)

1. In small plants where additional production facilities are available and due unsold product their prevails stock of product ready for dispatch as & when demanded, than breakdown maintenance is considered advantageous.
2. In small plants it is advantageous for non-critical machines, as the down time of such machines have no adverse effect on production.
3. Where the production activities can be quickly switch over on stand-by unit breakdown maintenance is advantageous.
4. In the units where absolute processes are in use and needs modern equipments for modernization of the processes, than for old processes breakdown maintenance is advantageous.



### 1.3.2.2 Dis-advantages of breakdown maintenance :

1. As it is impossible to know the time & location of breakdown in advance the planning of breakdown repairs cannot be done.
2. Breakdown machine is repaired in hastiness resulting into poor workmanship which leads to further breakdown.
3. It hampers the production resulting into the loss of production.
4. Machine stops during functioning in production, which leads to boredom and tiredness of the worker reducing their efficiency.
5. Fast deterioration of the plant & machineries.
6. Machine stops during work in progress, spoils the material and increases the waste. It also affects the quality control.
7. Possibility of frequent accident reduces the safety of workers & machines.
8. Accidental breakdown increases the production time hence overtime payment to the worker becomes necessary.
9. Sometime as a result of breakdown huge work of repairing arises which results into increased cost.
10. This system needs more stand by machines due to that capital investment in the plant increases.
11. There is no fix time to carry out the repair, hence workers are remaining jobless till the repairing is completed which spoils the industrial relations.
12. Machines whose functioning is regulated by the laws cannot be brought under the system of breakdown maintenance. Such machines are boiler, pressure vessels, lift, hoist, storage tank compressors.
13. In short till the machine breakdown, no care is exercised to keep machine in efficient condition, day by day quality of products is becoming poor, results into increased spoilage, wastage and finally the loss of market and customers, due to increased rejection.

### 1.3.3 Corrective maintenance :

(15)

Corrective maintenance is planned with a view to reduce breakdown, machines, units and parts of the units if found failing repeatedly due to same defect, then units & their parts are redesigned and used to prevent the breakdown. New design of machine makes it tough & strong due to that breakdown repair cost decreases over & above machines become safe in operation. To reduce the machine downtime engineering analysis is done and based on that new design for the machine is prepared. Parts are made stronger & wearing surfaces of the parts are made wear resistant. Machines are corrected in the sense of their construction. As the machine is becoming strong, the target of production can be achieved very easily cost of production can be controlled. The machines modified by corrective maintenance can give high quality products and hence the wastage, rejection etc. are reducing. Following are some of the work carried out in corrective maintenance :



**1.3.4 Scheduled maintenance :**

(3)

In this type of maintenance the work planning is done in such a way that the most of the maintenance job can be done on weekly holiday. Over and above to carry out maintenance of number of machines equal to the spare machines, the planning is done to carry out the work on working days. By planning the maintenance activities in above stated both the ways the machines which are engaged for production work are not required to stop only for to carry out the maintenance work. In this system some-time the faulty machines are used eventhough they are faulty till the weekly holiday. If such faulty machine met with breakdown before they are attended than the costly maintenance work is invariably arise. The use of faulty machinerie is adversely affects the quality of products & hence there are chances of increased rejection.

**1.3.5 Planned maintenance :**

In this type of maintenance the care is required to be taken for planning & record keeping of the maintenance work. The work like small adjustment, part repairing & replacement are done timely, considering their importance for efficient running of the machines. Here fault finding is done at right time as well as required repairing is also carried out to maintain such condition of the machine where chances of their metting with breakdown does not arise. In this type inspection, lubrication and adjustment etc. are done according to the plan and hence this type of maintenance is known as planned maintenance.

**1.3.6 Productive maintenance :**

In this type of maintenance the total attention for maintenance is concentrated on the important and most needed machines. The cost of maintenance of the unnecessary machines are being saved with a view to decrease the overall cost of maintenance in the plant by incurring the cost for only important & needy machines. This type of thinking overlook the not maintenance need of most of the machines in the plant. The condition of machines which are maintained due to this thought only are becoming more & more poor in their condition. The quality of products therefore made by such poor machines is also reduces. The work remaining pending on such machines are therefore transferred on important machines. This transfer of work results into increased load and hence important machines are overloaded. The running of important machines with such overload spoils their condition. Due to this reason the service life of important machines reduces and they are to be scrapped before the completion of their service life. Day by day the condition of machine shop becoming very very poor and a time will come at which huge capital replacement cast is required to be spent in the plant.

**1.3.7 Preventive maintenance :**

(13)

In this type of maintenance activities like inspection, lubrication, adjustments, minor and major repairing and complete overhauling etc. are done according to the prior planning of work. The charts showing various work to be done in each of the above activities are prepared in advance. Therefore based on such chart the machines are maintained at right



time. The fact "Prevention is better than cure" is always kept in mind in this type of maintenance. If there is fault in any machine, then it is removed when it is small in nature by proper repairing. This type of maintenance is the continuous attempt to keep the machines always in their good and efficient condition. Preventive maintenance is done on both running machines and if required by stopping the machine. The lubrication and adjustments are considered most important in preventive maintenance and hence carry out regularly & precisely. Fault finding and removal activities are done at right time in this preventive maintenance, therefore it offers an advantage of preventing the breakdown of machines. The preventive maintenance thus reduces the down time of the machines as well as the cost of maintenance. The preventive maintenance can be done by two methods; i.e.

1. Periodic maintenance
2. Routine maintenance.

#### **1.3.7.1 Periodic maintenance :**

This type of maintenance carried out as per prior planning considering production schedule and sales planning is called the "Periodic maintenance." The following activities are included in this type of maintenance :

1. Cleaning of the parts.
2. Oiling, greasing plus topping up & change of lubricants.
3. Checking the accuracies of precision machines like lapping machine, honing machine, boring machine etc.
4. Inspection of machineries as per schedules.
5. Testing of electrical equipments.
6. Adjustments of clearances.

Life of machine increases due to periodic maintenance.

#### **1.3.7.2 Routine maintenance :**

The daily carried out maintenance work like inspection, cleaning, lubrication, fitting & adjustments etc. are called routine maintenance. In this maintenance loose nut-bolts are tightened, replacement of unserviceable lamps, tubes and other electrical appliances etc. are done. On completion of production work on machine cleaning, lubrication, checking the lubricant level & functioning of lubricating oil pump are checked. In short the attention is given to the machines to maintain their good & efficient condition.

Routine maintenance is carried before starting the use of the machine at suitable time between the shift by keeping in mind that the time of production should be wasted for such work. This work is also undertaken during lunch hours.

#### **1.3.8 Predictive maintenance :**

By using the knowledge related to maintenance field and using sensitive instruments like vibration analyzer, amplitude meter, pressure, temperature & resistance strain gauges, the faulty conditions which may spoil the machine are found out. The faults are traced before machine stops functioning and repairing & replacement work is done alongwith adjustments



required to prevent the machine breakdown. The good & efficient condition of machine is obtained, the products quality improves in this predictive maintenance. The work of planning and record keeping is also reduces considerably.

This type of maintenance is modern approach, in which the factors which are responsible to make the machine faulty are continuously searched by using precision instruments. This modern work is carried out by the skilled workers. This latest technique based work gathers the needy informations continuously for the machines, units and parts as regards to their conditions. As a result of which this maintenance proves of utmost importance. The modern approach to supervise the condition of a machine and machine shop as a whole is developing day by day by the combine efforts and use of skilled workers and precision instruments. As a result of which new remedial measures are made available, and hence modern maintenance as per the need of modern time is done by using the approach of predictive maintenance.

### 1.3.9 Modular maintenance :

In this type of maintenance machines are modified as per the need of production. The store is also modified to fulfil the requirement of machine repairing work. The specifications of the machines are changed as a result of modifications eventhough care is exercised to avoid any greater change in comparison to the original specification of the machine. The central idea is to obtain quality products, prevention of wastage and economical growth of the industry. Engineering analysis is done to reduce maintenance time & cost reduction, machines, plant and processes are made to satisfy the production requirements. This type of maintenance is only giving importance to the production and hence it is a factor which proves its ability to achieve the target of production. Due to its capacity to achieve the production goal, right quantity of products with right quality at right time can be produced which helps in achieving sales target and income of the plant.

### 1.4 Types and application of tools used for maintenance : (14)

Following tools used for the maintenance work :

- (1) **Spanner** : There are four types (1) Plate Spanner (2) Ring Spanner (3) Box Spanner and (4) Adjustable Spanner.  
Plate, Ring and box spanners are fixed spanners and they are single or double ended used for fixed size bolt or nut head. Various size spanners are used for different size bolts. Adjustable Spanner is used commonly for various sizes in a range. Plate Spanner is used for bolt heads above the surface, Ring Spanners are used for bolt heads under shallow heads and box spanners are used for deep depth bolt heads.
- (2) **Allen Key** used for fixed sized slot on bolt heads. Various sizes are used for different size slots. Bolt head can be kept under surface by such design of bolt head. Allen key rotates bolt head by properly fitting in a slot provided on bolt head.
- (3) **Mallet** : It is made from wood or hard plastic and used for light blow on part surface to make fitting slightly loose.



- (4) Hammer : Used for light blow and hammering on the part surface, it is made from alloy steel and fitted with wooden handle.
- (5) Brush : Used to remove old paint grease and oil deposits. It is made of steel wire and wooden handle. Wires are fitted on wooden surface.
- (6) Emery Paper : Used for cleaning the surface in abrasive method.
- (7) Pump Jet : Used to generate jet of water by pressure, and water and air mixed jet is used for mechanical washing.
- (8) Blow Torch : Used for thermal cleaning of the surface under inspection. It provides blue flame for heating purpose.
- (9) Drift Pin : Used for cushioning effect in between surface and hammer during blowing.
- (10) Screw Driver : Used to drive screws for fitting and un-fitting. It is fitted on the slots provided on screw head.
- (11) Pipe Wrench : Used to fit and unfit pipe in a coupling for it.
- (12) Pliers : Simple as well as Slip Jaw pliers are used for pulling screws and nails from the joint.
- (13) Bearing Puller : Bearing is dismantled from it's housing by bearing puller.
- (14) Grease Gun : It is used for greasing the parts purpose.
- (15) Hand Oiler : Used to provide oil between rubbing surface when machine is in operation.
- (16) Straight Edge & Filler Gauge : Both are used jointly for measuring wear on plain surface.
- (17) Spirit Level : It is used to check straightness and squareness on machine surface like bed of the lathe machine.
- (18) Dial indicator and auto colimeter : Used for flatness and squareness of worn out surfaces.
- (19) Measuring Instruments : used for actual measurements of worn out surfaces. They are caliper micrometers. (Out side and Inside). Depth micrometer, screw thread micrometer, telescopic gauge, Bore gauge, Bevel protector angular gauge etc.

## **1.5 Maintenance Cost and its Relationship with Replacement Economy :**

### **1.5.1 Definition of maintenance cost :**

Maintenance cost is the cost incurred not for to increase the production capacity and accepted service life of the equipments but it is the cost of maintaining the equipments in its efficient condition.

The expenses mentioned below are included in the cost of maintenance :

1. Down – time cost
2. Cost of fault finding
3. Dismantling cost



4. Repair and maintenance cost
5. Material cost
6. Cost of maintaining & distributing the utility services.
7. Overhead cost of maintenance department.

It is very very issential to reduce the cost of maintenance as its reduction results into creased productivity.

All the direct and indirect expenses are included in the cost of maintenance. These are bour cost, supervision cost, store-keeping cost etc. are accounted in the overhead cost of e maintenance department.

### 5.2 Factors affecting the maintenance cost : (5/

The following are the factors which are affecting the maintenance cost:

1. Frequency of inspection, lubrication and repairing
2. Standardisation of parts and machines.
3. Excess stock of the replacement parts.
4. Excess maintenance staff.
5. Insufficient inspection of maintenance job.
6. Time interval between successive maintenance.
7. Design and maintainability of equipments
8. Inspection types and their number in use.
9. Skill of the maintenance staff.
10. Method of repairing
11. Payment of overtime.
12. Utilisation of uneconomical materials for repairing.

### 5.3 Factors for deciding replacement instead of maintenance :

The factors which are considered to decide replacement instead of maintenance are also called the causes of replacement. These are as follow :

1. Deteriorated machine
2. Machine obsolescence
3. Machine inadequency
4. Working condition and morale.

#### 5.3.1 Deteriorated Machine :

When the condition of machine is such that the targeted production is not achieved. Machine stops functioning due to the breakage, wearing of parts and other defects like change or alteration of the alignments. When it is felt that the présent condition of the machine is deteriorated very much as compared with its condition when machine was new than the machine should be replaced. This type of replacement is advantageous both for the production capacity and quality of products.



### 1.5.3.2 Machine obsolescence :

The development of Science and Technology has made the automatic machines available for replacing the present machines used in the production-shop. The newly available machines are capable of giving higher production, better quality & reduced cost of production. Hence the present machines are considered obsolete in comparison to the modern machines therefore it becomes necessary to replace the present machines. The replacement is also provide the capacity to the industry to survey in the present day competitive market.

### 1.5.3.3 Machine inadequency :

The development in machine building, in this days, resulted into the time of special purpose machines. The present machines in comparison to special purpose machines are giving lesser production capacity. The newer products can be produced only by using the special purpose machines. Therefore the inadequency is created in present machines due to invention of special purpose machines. This inadequency can only be removed by replacing the present machine by special purpose machines.

### 1.5.3.4 Working condition and morale :

If the use of any machine brings unsafe working condition then it should be replaced. The machine becoming hot, oil leakages developed, smoke emitted from the machine, machine produces excessive noise etc. are the unsafe conditions. Using the machine having such unsafe condition reduces the efficiency of the operators. The unsafe condition of machines and its effects on the operator are the causes of accidents, therefore such machines should be replaced.

### ✓ 1.5.4 Methods for selection of replacement : (1)

The following methods are used to decide when the machine can be replaced :

1. Annual cost method of replacement
2. Mapi method of replacement
3. Total life average method of replacement
4. Rate of return method of replacement
5. Present worth method of replacement.

The value of machine reduces with the increase in its life. The reduction in the value of machine is known as depreciation.

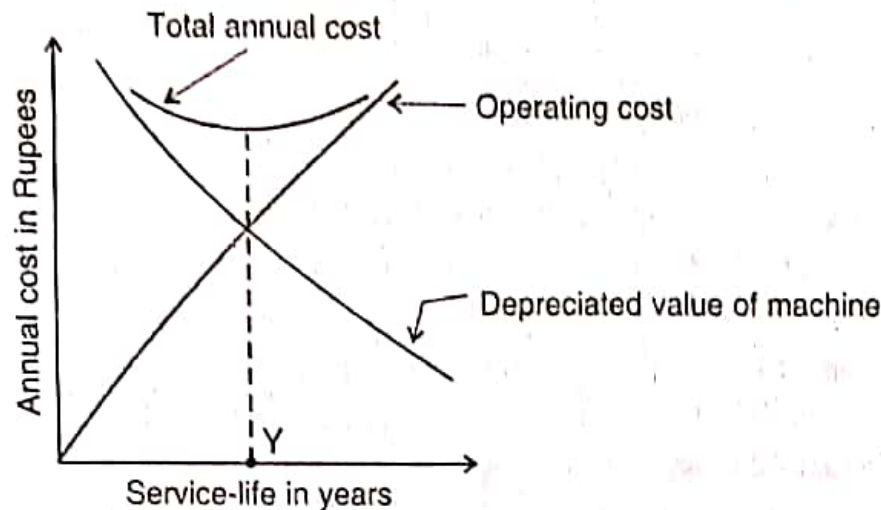
#### 1.5.4.1 Annual cost method of replacement :

With the increased life of machine its depreciation increases. In this method of replacement the depreciation is decided on annual basis. At the beginning of each year the depreciation is deducted from the cost of machine and cost balance of the machine is calculated. The depreciation is considered as equal to the loss of interest on the capital cost of the machine. The cost of maintenance, repairing, replacement & power etc. are called the operating cost increase in proportion to the increase in life of the machine. By adding the loss



of interest and operating cost the annual cost for operating the machine for one year can be found out.

The operating cost is calculated on the basis of cost statements of the maintenance department. In this operating cost the loss of interest for one year on capital investment on particular machine is added to calculate the annual cost. A graph showing the salvage price, operating cost and total annual cost against the service life of a machines is given in fig. 1.1.



(Fig 1.1: Cost V/S service life)

In the fig 1.1. the point 'Y' represents the life of machine in years at which the machine is to be replaced.

With the help of the above graph when the machine is to be replaced can be decided. When the total annual cost is minimum at that time operating cost & depreciated value of the machine both will be equal. The point - y in the graph represents the life of machine at that time. If the machine will be used beyond this life then its operating cost will increase. Therefore when machine completes its life of y years then it should be replaced. This type of help in decision making is provided by this graph. Moreover the graph itself is prepared on the basis of cost statements of the maintenance department and hence it can be said that the decision of replacement in this annual cost of replacement is decided on the basis of the data available in maintenance department.

#### 1.5.4.2 Mapi method of replacement :-

(7)

The capital investment of each machine and equipment can be recovered in certain fixed years. The capital cost is recovered by the saving of depreciation. Machinery & allied products institute is abbreviated as "MAPI". The MAPI has given a quantitative approach for deciding replacement. When machine will become old then it will become obsolete is the principle adopted in this method. By applying this principle the economical service-life of the machine is calculated as under :

Let  $G$  = Inferiority gradient proportional to the life of the machine

$N$  = Retentive period or Economical service life of the machine.

$I$  = Interest rate



S = Salvage value

C = Capital cost or purchase cost.

F = Annual expenses of first year.

$$\text{Economical service life, } N = \frac{2(C-S)}{G}$$

By calculating the value of 'N', we can decide when the machine should be replaced. At the time of replacement total cost can be given by the formula.

$$\text{Total cost} = \frac{G(N-1)}{2} + F + \frac{C-S}{N} + \frac{i(C+S)}{2}$$

The time is the important factor for capital investment in machinery, equipment and method. By using the present or defending machine for one year more, how much cost will increase. If replacement is done earlier, then the operating cost will increase. The MAPI method is comparing the cost of using present or defending machine for using one year more and cost of new machine or challenger. The machine purchased today needs replacement after some year due to the development of technology. MAPI method is giving exact information for the future. Till the total cost of machine remains minimum, it is used and thereafter a decision for replacement is taken. Inferiority gradient is the rate of obsolescence and deterioration. Due to the combine effect of obsolescence & deterioration, increase in operating cost is noticed.

#### 1.5.4.3 Total life average method of replacement :

In this method of replacement capital cost & operating cost both are added and the sum of them is divided by the approximate life of the machine, to find the average annual cost  
Total cost = capital cost + operating cost + interest

$$\text{Average cost per year} = \frac{\text{Total cost}}{\text{Approximate life}}$$

The average cost per annum is less when the machine is new. But when the machine becomes old, its annual average cost increases and hence the decision for replacement can be taken.

#### 1.5.4.4 Rate of return method of replacement :

In this method how much return is given by the machine is taken into consideration. In this term return the output of machine, quality of products, operational ease, machine maintenance cost etc. are included. New machines are giving more return but due to increase in repairing when the machine is becoming older the rate of return decreases. Therefore base on the reduced rate of return, machine replacement time is decided.



#### 1.5.4.5 Present worth method :

With the development of technology the method of production is becoming obsolete. Therefore replacement decision is taken to replace the machine by a machine produced as per latest technology. e.g. Developed industries are replacing older lathe by CNC lathe.

#### 1.5.5 Factors considered to decide replacement :

The following factors are considered while making decision regarding replacement :

- |                            |                                |
|----------------------------|--------------------------------|
| 1. Maintenance cost        | 2. Reliability index           |
| 3. Latest Technology       | 4. Increase in production      |
| 5. The cost of new machine | 6. Amortization period         |
| 7. Bath tub curve          | 8. Repairing facility          |
| 9. Quality after repairing | 10. Operating cost of defender |

### 1.6 Service-Life of Equipments & Machines :

#### 1.6.1 Definition of service life :

If the machines & equipments are used in mass production, the depreciation is charged at a rate of 15% and if used for unit production the rate of depreciation is 10% considered. The depreciated amount is calculated as per the applicable rate and then deducted from the capital cost of the machines and equipment. The cost of new parts used in machine during repairing is added in the cost so calculated. The number of years of life of machine required to recover the capital cost is known as the service-life of the machines & equipments. In short the service-life is the period between the two replacement.

#### 1.6.2 Factors prolonging the service life :

The factors which are considered to prolong the service life of the machines and equipments are as under :

1. Proper use and care of the machines
2. Action minimize the friction
3. Improved lubrication
4. Use of wear compensating elements.

##### 1.6.2.1 Proper care and use of the machines :

1. Use the machines within its specified capacity.
2. Regular cleaning and adjustment
3. Apply grease and oil as per lubrication chart at regular intervals.
4. Repair the machine by finding fault at right time.
5. Use always standard parts for replacement
6. Use safe processes.

##### 1.6.2.2 Action to minimize the friction :

1. Sufficiently lubricate the wearing surfaces.
2. Provide new & repaired parts a good surface finishing.

3. Provide wear resistant coating on the wearing surfaces of the parts.
4. The hardness of the working surfaces should be of desired hardness index.
5. Save the wearing surfaces from dust, dirt and metal chips.

#### 1.6.2.3 Improved lubrication :

1. Always use proper grades of lubricants for the machine & its parts lubrication as per the recommendations of the manufacturer.
2. Lubrication should be done at proper time.
3. Sufficient lubrication should be applied.
4. Prepare lubricating programmes like daily, weekly etc. and implement the programmes at right time.
5. Change the lubricating oil of the reservoir in each unit of the machine as per the change programme.

#### 1.6.2.4 Use of wear compensating units : (9)

The elements like bushings, taper key and lines etc. can compensate the wear of the parts. These wear compensating units are of two types :

1. The bush, liners etc. are the fixed type elements and
2. Wedges, taper sleeve and taper key are the adjustable elements.

#### 1.7 Factors Affecting the Service Life of the Equipments : (13)

The following are the factors which are influencing the service life of the machines and equipments :

1. Intensity of friction
2. Quality of parts material
3. Type of lubricant and lubricating method
4. The dirt & dust present in the atmosphere.
5. Friction prevailing between the wearing surfaces.
6. The magnitude and type of load carried by the parts such as normal load or cyclic load.
7. High temperature
8. Surface finish of the wearing surfaces of the parts.
9. Type of machines & equipments such as precision or non-precision and critical or noncritical.
10. Temperature and its variation in the plant.
11. The type of maintenance used in the plant.
12. Surface hardness & smoothness of the parts as hard & smooth surface are experienced lesser wear.

#### EXERCISE

1. Explain the term maintenance.
2. State the aims of maintenance department.
3. State the primary & secondary functions of maintenance department.



4. List the primary & secondary functions of maintenance department and explain any one from each category.
5. Write short notes on :
  - (i) Plant protection (ii) salvage (iii) plant waste disposal and
  - (iv) creation & sell of utility services.
6. List responsibility of maintenance department.
7. State the types of maintenance & explain who is responsible for each type.
8. Write short notes on :
  - (i) Capital replacement, (ii) Breakdown maintenance and (iii) Preventive maintenance.
9. Define the term maintenance cost.
10. Explain the need of maintenance in production industry.
11. Show the organization structure of maintenance department of large scale industry.
12. State the factors affecting maintenance cost.
13. State and explain the factors to be considered to decide replacement instead of maintenance.
14. Explain MAPI – Method for replacement.
15. Define the service life of equipment.
16. State & explain the factors to be considered to prolong the service life of the equipment.
17. Define the maintenance cost the state the factors affecting it.
18. Explain briefly MAPI principle as used in replacement decision.
19. State the types of industrial maintenance and explain in which condition each type is used.
20. Write the factors affecting service life of equipment.
21. Explain any two :
  - (i) Periodic Inspection (ii) Cleaning method (iii) Breakdown maintenance.
22. Explain briefly relationship between maintenance & production department.
23. Differentiate between preventive maintenance & breakdown maintenance.
24. State atleast four objectives of maintainability & explain any one of them.
25. Define and explain maintainability.

